

# Syllabus

## Unit-2

**Teaching Hours: 12**

### **DATA PROCESSING USING PYTHON**

#### **Structured Dataset Handling**

Dataset ingestion from structured sources – tabular data representation – integration of data from multiple structured inputs – dataset formatting for processing workflows.

#### **Data Transformation and Preparation**

Data filtering, aggregation, merging, and reshaping in workflow context – data normalization and standardization – data validation and consistency checks – preparing curated datasets for storage and analytical systems.

# Syllabus

## Text & Reference Books

- [1] Fundamentals of Data Engineering, Joe Reis and Matt Housley, O'Reilly Media, 2022.
- [2] Designing Data-Intensive Applications, Martin Kleppmann, O'Reilly Media, 2017.
- [3] Digital Computer Fundamentals, Floyd, Thomas L, Pearson International, 11th Edition, 2015

## Essential Reading

- [1] Data Engineering with Python, Paul Crickard, Packt Publishing, 2020.

# Learning Direction: The Data Engineering Lifecycle

## 1. Collection & Ingestion

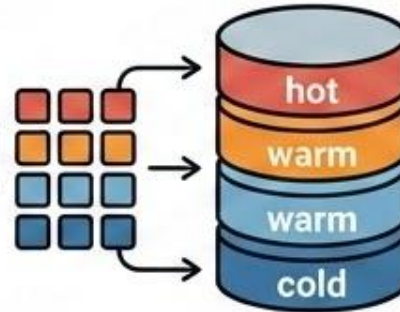


## 2. Cleaning & Transformation



How it is cleaned  
and transformed

## 3. Efficient Storage



How it is stored  
efficiently

## 4. Analytics & AI



How it is used for  
analytics and AI

## 5. Security, Archival, & Deletion



How it is secured,  
archived, and deleted

**These stages form the core of our learning journey in Data Engineering.**

# Structured Dataset

## STRUCTURED DATASETS

Organized, Fixed Format, Easy to Store & Query

### WHAT IT LOOKS LIKE



| ID  | Name    | Age | City      | Salary |
|-----|---------|-----|-----------|--------|
| 101 | Alice   | 28  | Mumbai    | 60000  |
| 102 | Bob     | 34  | Bangalore | 75000  |
| 103 | Charlie | 26  | Delhi     | 48000  |
| 104 | Diana   | 30  | Hyderabad | 62000  |

### EXAMPLES



Excel  
Sheets



SQL  
Databases



CRM  
Data



Sales  
Records



Bank  
Transactions

### REAL APPLICATIONS / USE CASES

- ✓ Financial reporting & dashboards
- ✓ Inventory management
- ✓ Customer data management
- ✓ Business intelligence & analytics
- ✓ Quick filtering, sorting & reporting

### KEY STRENGTHS

- ✓ Easy to store in databases
- ✓ Fast query and retrieval
- ✓ Well-defined schema
- ✓ High accuracy and consistency



VS

## UNSTRUCTURED DATASETS

No Fixed Format, Harder to Store & Analyze

### WHAT IT LOOKS LIKE



Emails



Documents  
(PDF, Word)



Images



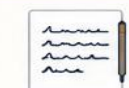
Videos



Social Media  
Posts



Audio  
Recordings



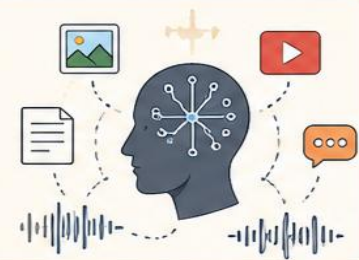
Handwritten  
Notes



Chats &  
Messages

### REAL APPLICATIONS / USE CASES

- ✓ Sentiment analysis from social media
- ✓ Image recognition (medical, security)
- ✓ Voice to text & speech analytics
- ✓ Document understanding & text extraction
- ✓ Chatbot support from emails/chats
- ✓ Fraud detection from unstructured logs



### CHALLENGES OF UNSTRUCTURED DATA



- ✗ No fixed structure or schema
- ✗ Hard to store and organize
- ✗ Difficult to search and retrieve
- ✗ Requires advanced tools (NLP, CV, ML)
- ✗ High processing time and cost
- ✗ Data quality and inconsistency
- ✗ Harder to ensure security and compliance



### KEY TAKEAWAY:

Structured data is easy to manage and analyze with traditional tools.  
Unstructured data is complex but holds deep insights when analyzed with AI/ML.

# What is Data Ingestion?

Structured Dataset Handling

in  
fo

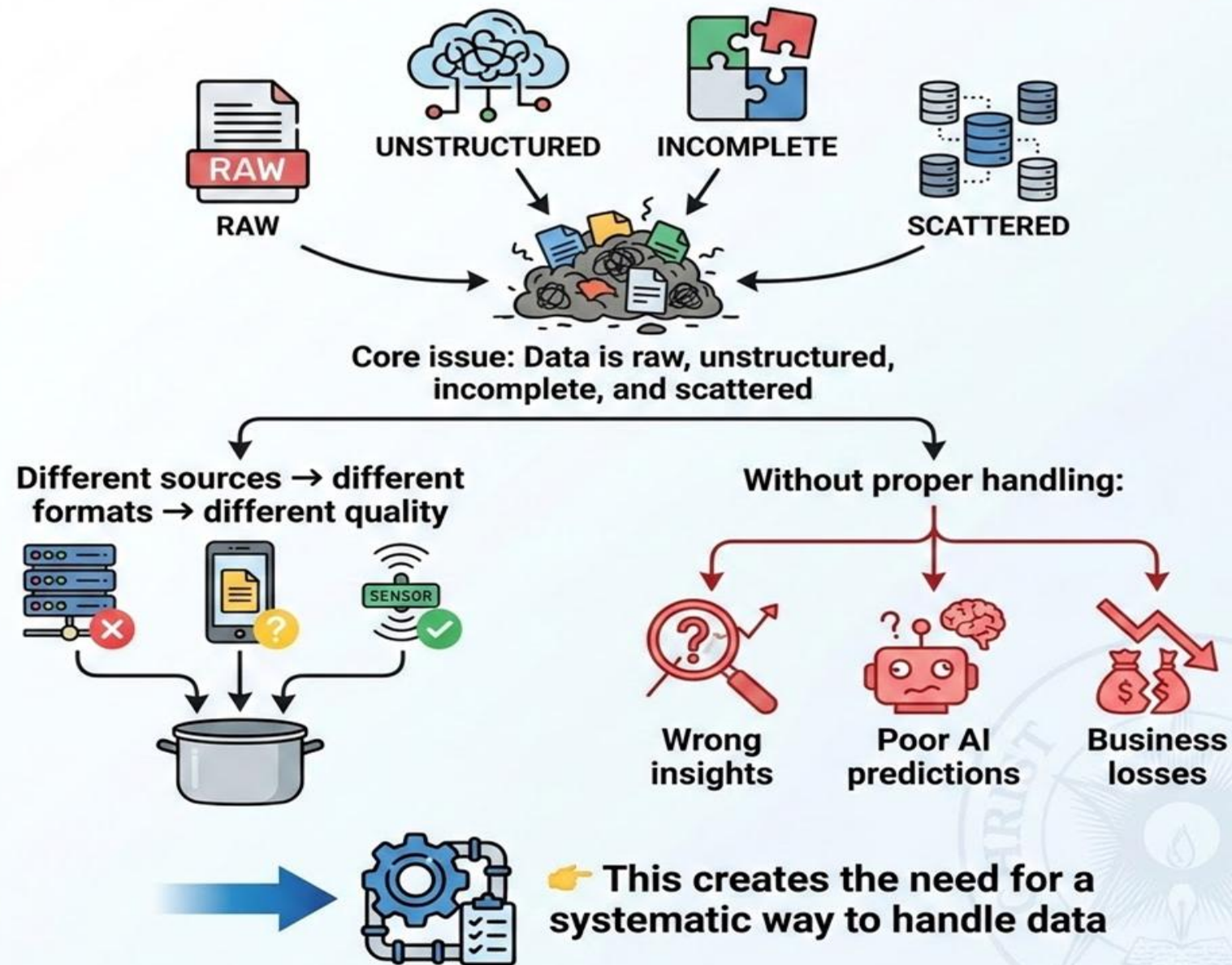
**Data Ingestion** is the process of collecting data from one or more sources and bringing it into a system where it can be **stored, processed, and analyzed**.

## The Ingestion Equation

Data Ingestion = **Taking** data from source + **Loading** into system

(Database, Data Warehouse, Data Lake, or Analytics System)

# Problem / Challenge



# Step 1: Data Collection

The first step in data ingestion: gathering diverse information from multiple sources.

## storage

### Databases

#### Structured Data

Relational databases like SQL Server, MySQL, and Oracle.

#### Example:

*Customer info, sales, inventory.*

## description

### Files

#### Semi-Structured

Sourced from log files, CSV, JSON, and XML formats.

#### Example:

*Web logs, config files, datasets.*

## api

### APIs

#### Web Services

Data fetched through RESTful APIs and service protocols.

**Example:** *Social media, weather, markets.*

## sensors\_kr

### Streaming

#### Real-Time Data

Platforms like Apache Kafka, Kinesis, and Event Hubs.

**Example:** *Stock tickers, live feeds, sensor streams.*

## router

### IoT Devices

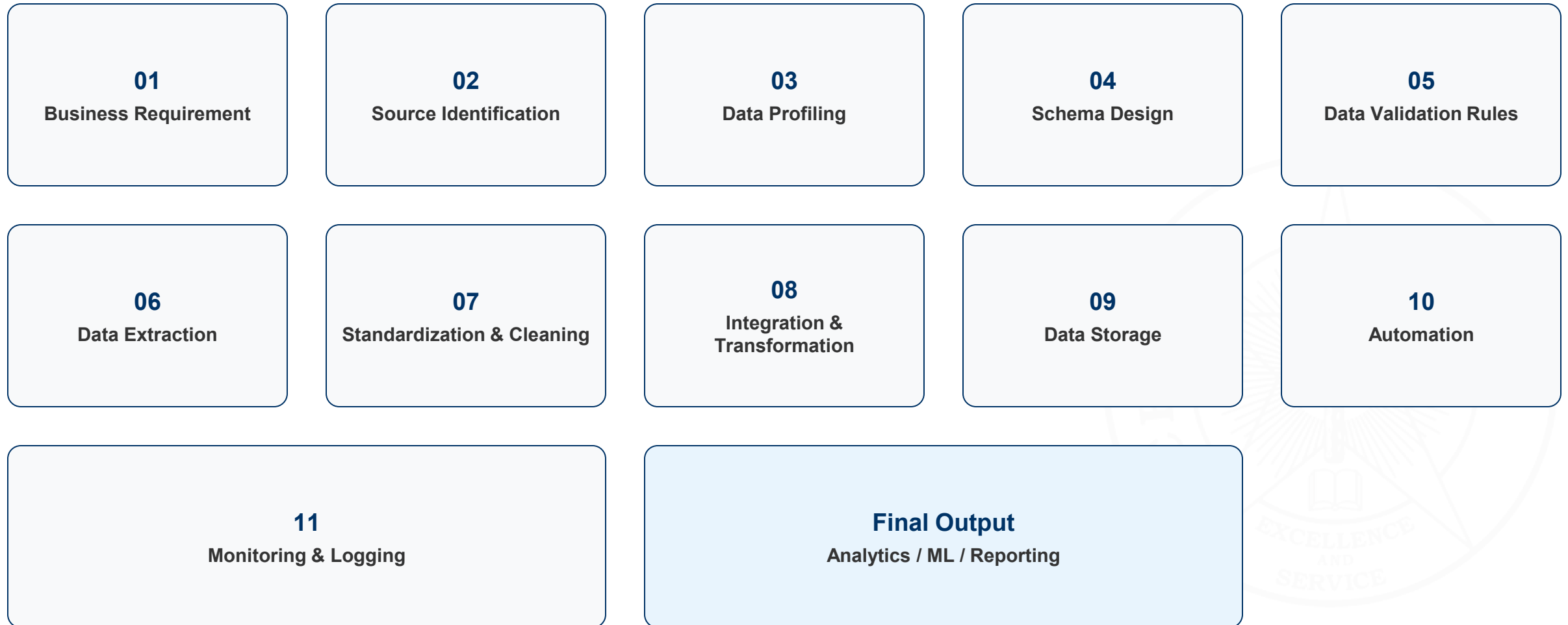
#### Sensor Data

Information from smart devices and industrial sensors.

#### Example:

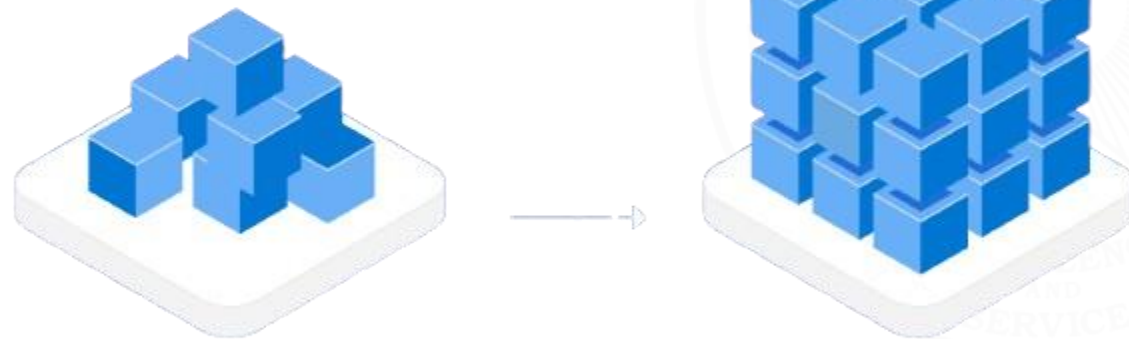
*Temperature logs, home automation.*

# Workflow of Data Ingestion



# Data Transformation and Preparation

Data Transformation and Preparation is the process of converting raw, inconsistent, and heterogeneous data into a structured, clean, and analytics-ready format suitable for data warehouses, dashboards, business intelligence systems, and machine learning models.



# Data Transformation and Preparation

## **zomato** Industry Use Case: Zomato Food Delivery Platform



Raw Data Received

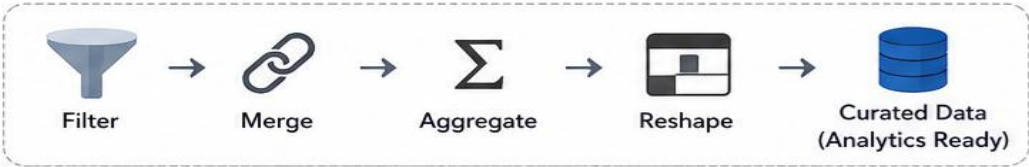
**Orders Table**

| Order_ID | Customer | Restaurant  | City      | Amount | Status    |
|----------|----------|-------------|-----------|--------|-----------|
| 101      | Rahul    | Pizza Hut   | Bangalore | 450    | Delivered |
| 102      | Priya    | Dominos     | Bengaluru | 650    | Delivered |
| 103      | Ravi     | Pizza Hut   | BLR       | 520    | Cancelled |
| 104      | Asha     | Burger King | Bangalore | NULL   | Delivered |

**Notice the problems:**

- Bangalore written in 3 ways (Bangalore, Bengaluru, BLR)
- Missing amount (NULL)
- Cancelled order (Status = Cancelled)
- Different restaurants (Pizza Hut, Dominos, Burger King)

**This is raw data** – it contains inconsistencies and issues. It needs to go through cleaning, validation and transformation before it can be used for analytics.



# 1 Data Filtering



## Meaning

Filtering means keeping only the records we need.

*Think of it as using a sieve.*



## Business Requirement

The Finance team wants only successful orders.

### RAW ORDERS DATA

| Order_ID | Customer | Restaurant  | City      | Amount | Status    |
|----------|----------|-------------|-----------|--------|-----------|
| 101      | Rahul    | Pizza Hut   | Bangalore | 450    | Delivered |
| 102      | Priya    | Dominos     | Bengaluru | 650    | Delivered |
| 103      | Ravi     | Pizza Hut   | BLR       | 520    | Cancelled |
| 104      | Asha     | Burger King | Bangalore | NULL   | Delivered |



```
SELECT *  
FROM Orders  
WHERE Status = 'Delivered';
```

### RESULT (After Filtering)

| Order_ID | Restaurant  | Amount |
|----------|-------------|--------|
| 101      | Pizza Hut   | 450    |
| 102      | Dominos     | 650    |
| 104      | Burger King | NULL   |

✓ The cancelled order is removed.

## Real Industry Examples



### Marketing Team wants

Only Bangalore customers



```
SELECT *  
FROM Orders  
WHERE City = 'Bangalore';
```



### Finance Team wants

Orders above ₹500



```
SELECT *  
FROM Orders  
WHERE Amount > 500;
```



### Operations Team wants

Today's orders only



```
SELECT *  
FROM Orders  
WHERE Order_Date = CURRENT_DATE;
```



Filtering reduces unnecessary data before processing.



Improves performance



Reduces data volume



Helps teams focus on what matters most

## 2 Data Aggregation

Summarizing data to get valuable insights



### Meaning

Aggregation means summarizing data. Instead of looking at every order individually, we calculate key metrics.

### We Calculate



#### Total Sales

Total revenue generated



#### Average Rating

Average customer rating



#### Total Orders

Total number of orders



#### Maximum Revenue

Highest revenue value

### EXAMPLE



#### Orders (Raw Data)

| Restaurant | Amount (₹) |
|------------|------------|
| Pizza Hut  | 450        |
| Pizza Hut  | 520        |
| Dominos    | 650        |



Turning many records into meaningful summary

### AGGREGATE



#### SQL Query

```
SELECT Restaurant,  
SUM(Amount)  
FROM Orders  
GROUP BY Restaurant;
```



We group the data by Restaurant and calculate the total amount.



### OUTPUT

#### Aggregated Result

| Restaurant | Total Sales (₹) |
|------------|-----------------|
| Pizza Hut  | 970             |
| Dominos    | 650             |



All sales for the same restaurant are added together.

### INDUSTRY USES



Manager asks:

*"Which restaurant earned the most today?"*



Data Engineer

creates aggregated tables to answer business questions quickly.



### Key Takeaway

Aggregation helps convert detailed raw data into **meaningful summaries** that drive better business decisions and faster reporting.



Better Insights



Faster Reports



Smarter Decisions



Business Impact

# 3 Data Merging

Combining data from multiple tables to get complete information



## Meaning

Different departments store different information.  
To answer business questions, we combine tables.  
This is called **Merging (Joining)**.



Orders Table

| Order_ID | Customer_ID | Restaurant_ID |
|----------|-------------|---------------|
| 101      | C01         | R10           |



Customer Table

| Customer_ID | Customer |
|-------------|----------|
| C01         | Rahul    |



Restaurant Table

| Restaurant_ID | Restaurant |
|---------------|------------|
| R10           | Pizza Hut  |

## MERGING (JOINING)



## AFTER MERGE

| Order | Customer | Restaurant |
|-------|----------|------------|
| 101   | Rahul    | Pizza Hut  |



Now we have **complete information** about the order!



## SQL

### SELECT

```
o.Order_ID,  
c.Customer,  
r.Restaurant
```

### FROM Orders o

### JOIN Customer c

```
ON o.Customer_ID = c.Customer_ID
```

### JOIN Restaurant r

```
ON o.Restaurant_ID = r.Restaurant_ID;
```



Orders  
(o)

JOIN



Customer  
(c)

JOIN



Restaurant  
(r)



## Industry Uses

Business wants to know

*"Which customers order from which restaurants?"*



Impossible without merging.



### Customer Analysis

Understand customer preferences and ordering patterns.



### Revenue Reports

Calculate total revenue by customer, restaurant, city, etc.



### Personalization

Recommend restaurants based on customer order history.



### Operational Insights

Track performance of restaurants and delivery partners.



## Key Takeaway

Merging (Joining) connects data from different tables



Creates a complete picture



Enables better analysis and decision making



Powerful for business intelligence!



## 4 Data Reshaping

Changing the structure of data without changing the information



### Meaning

Changing the structure of data. The information stays the same, only its layout changes.



### Why?

BI tools like the following often require reshaped data.



Power BI



Tableau



Excel



Reshaping makes data **easy** to analyze and visualize.

### ORIGINAL DATA (LONG FORMAT)

| Restaurant | Month | Sales (₹) |
|------------|-------|-----------|
| Pizza Hut  | Jan   | 12000     |
| Pizza Hut  | Feb   | 18000     |
| Pizza Hut  | Mar   | 15000     |



Data is stored in rows (one month per row)

↓  
This is called  
**PIVOTING**

### BUSINESS WANTS (WIDE FORMAT)

| Restaurant | Jan   | Feb   | Mar   |
|------------|-------|-------|-------|
| Pizza Hut  | 12000 | 18000 | 15000 |



Data is spread across columns (one month per column)

### OR THE OPPOSITE

#### Wide Table

|  |  |  |
|--|--|--|
|  |  |  |
|  |  |  |



**UNPIVOTING**  
(Reverse of Pivoting)



#### Long Table

|  |  |  |
|--|--|--|
|  |  |  |
|  |  |  |

Wide Table → Long Table  
is called **UNPIVOTING**

### COMPLETE ETL FLOW SO FAR



Raw Orders



**Filtering**

(Remove cancelled)



**Merging**

(Customer + Restaurant)



**Aggregation**

(Total Sales)



**Reshaping**

(Pivot by Month)



### EXAMPLE VISUAL: PIVOTING IN ACTION

#### LONG FORMAT (BEFORE PIVOT)

| Restaurant | Month | Sales (₹) |
|------------|-------|-----------|
| Pizza Hut  | Jan   | 12000     |
| Pizza Hut  | Feb   | 18000     |
| Pizza Hut  | Mar   | 15000     |

PIVOT  
→

#### WIDE FORMAT (AFTER PIVOT)

| Restaurant | Jan   | Feb   | Mar   |
|------------|-------|-------|-------|
| Pizza Hut  | 12000 | 18000 | 15000 |



### Key Takeaway

Reshaping (Pivoting/Unpivoting) helps convert data into the right structure for reporting, dashboards and analysis.

## 5

# Data Normalization

Making data consistent for accurate and reliable analysis



## Meaning

Normalization means making data **consistent**.

### Example 1: City Names



#### Raw Data

| City      |
|-----------|
| Bangalore |
| Bengaluru |
| BLR       |

All represent  
the same city.



#### Normalize to

| City      |
|-----------|
| Bengaluru |



### Example 2: Restaurant Names



#### Raw Data

| Restaurant |
|------------|
| Pizza Hut  |
| pizza hut  |
| PIZZA HUT  |



#### Normalize to

| Restaurant |
|------------|
| Pizza Hut  |



### Example 3: Phone Numbers



#### Raw Data

| Phone Number   |
|----------------|
| 9876543210     |
| +91-9876543210 |
| 09876543210    |



#### Normalize to

| Phone Number   |
|----------------|
| +91 9876543210 |



## Industry Benefit

Reports become

**ACCURATE.**

Otherwise,  
Bangalore sales appear as

- ✗ Bangalore
- ✗ Bengaluru
- ✗ BLR



which produces

**THREE SEPARATE ROWS**  
instead of one.



**Normalization ensures**  
one value is stored  
in one standard format.



## Why Normalize?



Accurate Reporting



Better Data Analysis



Avoids Duplicate Rows



Improves Performance



Ensures Data Quality

## 6

# Data Standardization

Convert data into **one common scale or format**



## Meaning

Convert data into **one common scale or format**.

Two common meanings in industry:



## A. Standardizing Formats

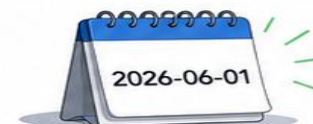


Dates

| Raw Data     |
|--------------|
| 01-06-26     |
| June 1, 2026 |
| 2026/06/01   |

Standardize to

| Standard Format |
|-----------------|
| 2026-06-01      |



All dates in the same YYYY-MM-DD format



Currency

| Raw Data |
|----------|
| 450 INR  |
| ₹450     |

Standardize to

| Standard Format |
|-----------------|
| 450             |



All currency in numeric format



## B. Standardizing Numerical Values (Machine Learning)



Original Data

| Delivery Time (minutes) |
|-------------------------|
| 15                      |
| 25                      |
| 40                      |
| 55                      |



After Standardization

| Standardized Values |
|---------------------|
| -1.20               |
| -0.30               |
| 0.75                |
| 1.50                |



Useful for ML algorithms



Standardization brings all values to a common scale with **mean = 0** and **standard deviation = 1**.



## Why Standardize?



Ensures consistency across systems



Improves data quality



Enhances accuracy in analysis



Essential for machine learning



Better reporting and decision making

# 7 Data Validation

Ensuring data is **accurate**, **complete** and **reliable**



**Meaning:** Checking whether data is **correct**.

## Example Rules



Order Amount

Must be

Amount > 0



Delivery Rating

Must be

1 – 5



Delivery Time

Cannot be

Negative

## Bad Data



Invalid Amount  
(Negative)

Order

101

Amount

-500



Rejected.

Missing  
Customer

Order

102

Customer

NULL



Flagged for  
correction.

## Industry Example



If a payment amount is

**₹ -800**

the pipeline **stops** and  
**alerts** engineers.



Data Ingestion



Validation Check



Validation Failed



Pipeline Stopped



Alert Engineers



**Why Data  
Validation  
Matters?**



Prevents bad data  
from entering the  
system



Improves accuracy  
of reports and  
dashboards



Ensures trust in  
data for analysis  
and decisions



Reduces operational  
issues and financial  
errors

# 8 Consistency Checks

Compare multiple datasets to ensure they **agree**.



## Example 1: Status vs Payment Consistency

### Orders Table

| Order_ID | Status    |
|----------|-----------|
| 101      | Delivered |



### Payment Table

| Order_ID | Payment |
|----------|---------|
| 101      | Failed  |



How can an order be **DELIVERED** if payment **FAILED**?



**Inconsistency**  
This is an **INCONSISTENCY**.



## Example 2: Amount Consistency

### Orders



Total Amount  
**₹500**



### Payment

Paid Amount  
**₹450**



**Mismatch**  
detected.



Industry pipelines  
generate **ALERTS**  
automatically.

## How Consistency Checks Work in Data Pipelines



Multiple Data  
Sources



Data Ingestion



Consistency  
Checks



Mismatch  
Found



Alert Generated



Engineers  
Investigate



Issue Resolved



Trusted Data



**Why Consistency  
Checks Matter?**



Ensures data  
integrity



Prevents wrong  
business decisions



Improves trust in  
data and reports



Detects errors  
early

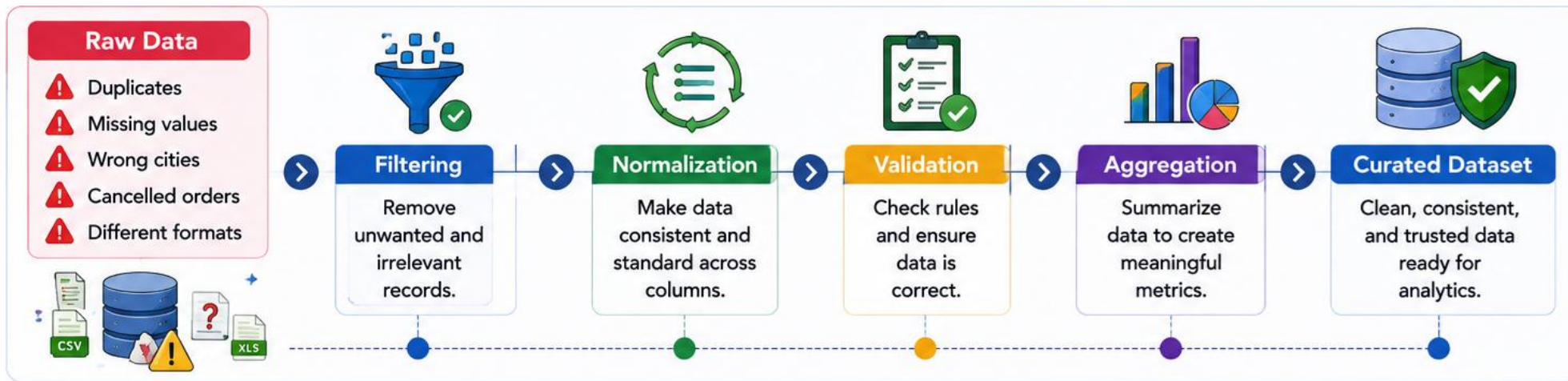


Reduces financial &  
operational risks



# 9 Preparing Curated Datasets

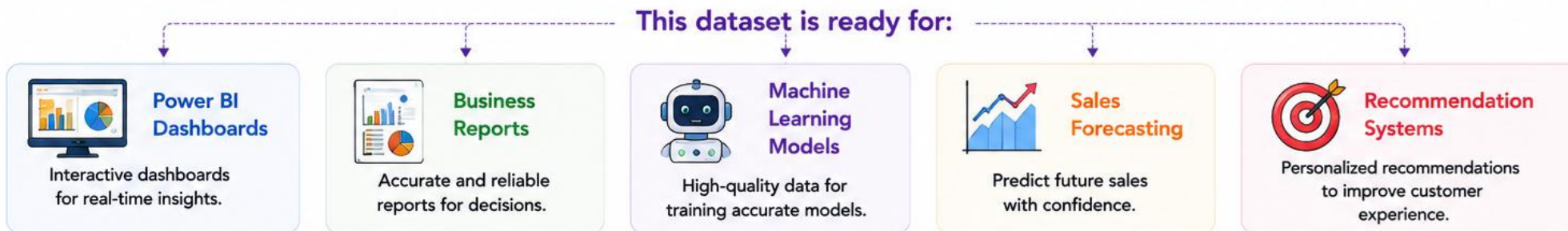
After all cleaning is complete, the data is transformed into a **trusted dataset**.



**Example**

| Date       | Restaurant | City      | Orders | Revenue   |
|------------|------------|-----------|--------|-----------|
| 2026-06-01 | Pizza Hut  | Bengaluru | 250    | ₹1,25,000 |

Now this dataset is **READY** for powerful analytics and AI.



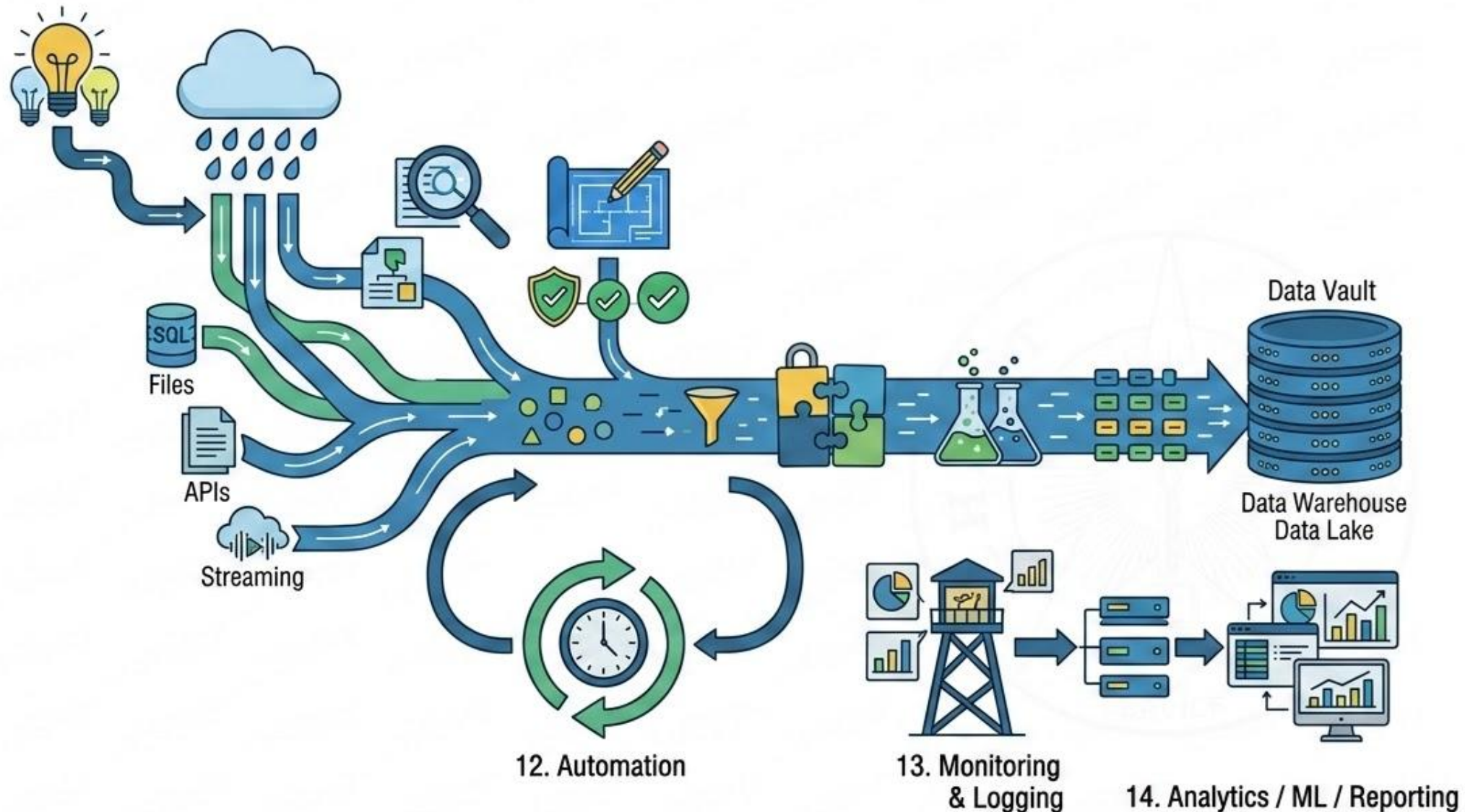
**Benefits of a Curated Dataset**

- Trusted and High-Quality Data
- Accurate Analytics
- Faster Decision Making
- Better Customer Experience
- Business Growth and Efficiency



# Workflow of Data Ingestion

1. Business Requirement
- ↓
2. Source Identification
- ↓
3. Data Profiling
- ↓
4. Schema Design
- ↓
5. Data Validation Rules
- ↓
6. Data Extraction
- ↓
7. Data Standardization
- ↓
8. Data Cleaning
- ↓
9. Data Integration
- ↓
10. Data Transformation
- ↓
11. Data Storage
- ↓
- Automation
- ↓
- Monitoring & Logging
- ↓
- Analytics / ML / Reporting



# Thank you

